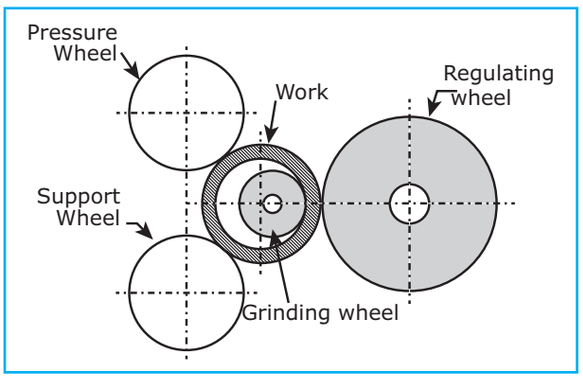
the contours of these two wheels is the wall thickness of the work.



Internal Centreless Grinding

Advantage of centreless grinding

1. The workpiece is supported throughout the entire length, grinding is done very accurately.
2. Small, fragile or slender workpieces can be grind easily.
3. Work holding devices such as chucks, dogs, centres, mandrels are not required.
4. The process is continuous, it is best adapted for mass production.
5. The size of the work can easily be controlled.
6. Less skilled worker is sufficient.

Disadvantages of centreless grinding

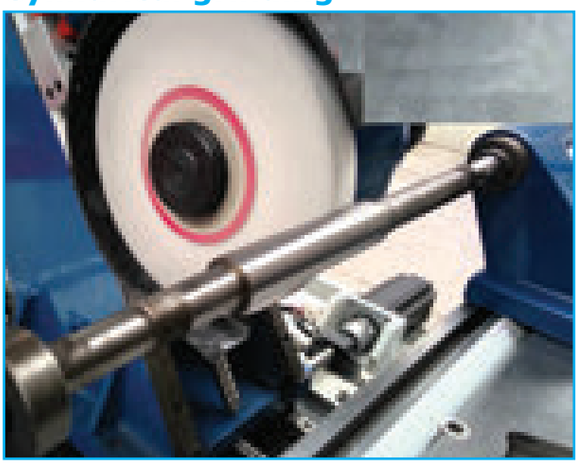
1. In hollow work, there is no certainty that the outer diameter will be concentric with the inside diameter.
2. Works having multiple diameters are not handled easily.

4.7 Grinding operations

Grinding is metal cutting operations of removing excess material from metal parts by means of a rotating hard abrasive wheel that act as a tool. The following operations are generally performed in a grinding machine.

1. Cylindrical grinding
2. Taper grinding
3. Gear grinding
4. Thread grinding

Cylindrical grinding



Cylindrical grinding is a metal cutting operation, where metal is cut by a rotating abrasive wheel in a cylindrical grinding is performed by mounting and rotating the work between two centres in a cylindrical grinding machine. The work is fed longitudinally against the rotating grinding wheel to perform grinding. The upper table of the grinding machine is set at 0^0 during the operation.

Taper Grinding

Taper grinding on long work pieces can be done by swiveling the upper table at required angle. If the work piece is short, the wheel hand may be swiveled by the taper angle. Another method of grinding external taper is to true the face of the grinding wheel by a diamond tool dresser to the required angle. In this case, the wheel head and the table are not swiveled.

XII BME_English version CHAPTER 04.indd 85 1/8/2020 1:43:06 PM